Roshan Rajendra Talathi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.44
Pimpri Chinchwad Polytechnic	Class XII (Maharashtra State Board Of Technical Education)	2015	79.52%
V.H.Paranjpe, Mahad	Class X (Maharashtra State Board of Secondary and Higher Secondary Education)	2012	73.64%

PROJECTS

M.Tech Thesis: Neural Network-Based Inverse Design for Soft Robotics

- Generating soft-robotic deformation data using a mathematical model in Python.
- Verifying the generated data against CAE simulation results.
- Training a neural network using the validated data for rapid inverse design.

Mini-Projects:

- **Machine Learning-Based Prediction of Iron Alloy Properties**

Developed a machine-learning pipeline to predict Ultimate Tensile Strength (UTS) and Elastic Modulus of iron alloys, involving data preprocessing, feature engineering, and model development using Linear Regression and ANN, with performance evaluated using MSE, RMSE, and R^2.

- **Bank Term Deposit Subscription Prediction using Random Forest**

Developed a Random Forest classification model to predict customer subscription to term deposits using demographic, financial, and marketing campaign attributes, with performance evaluated using classification metrics and a confusion matrix.

- **CIFAR-10 Image Classification using AlexNet**

Developed an AlexNet-based CNN model for CIFAR-10 image classification and optimised its performance through hyperparameter tuning, evaluating results using classification metrics and a confusion matrix.

- **ML-Based Inverse Design of Minimum-Weight Composite Laminates for Buckling Resistance**

Developed an ANN-based inverse-design framework to determine the minimum number of plies and suitable laminate configuration for carbon fibre composites subjected to compressive buckling using ANSYS.

WORK EXPERIENCE

Design Engineer

Auto Alpha Engineering Pvt. Ltd (Adient India Tech Centre), Pune

- Designed and modified automotive seat foam and head restraint components using CATIA V5, considering craftsmanship, comfort, packaging, and manufacturing requirements.
- Managed and maintained structured BOM and product data in Siemens Teamcenter, ensuring data accuracy and consistency across design revisions, component changes, and product configurations.
- Took ownership of design activities for European-region seating programs for global OEMs, including Volvo, Porsche, and Stellantis, coordinating with cross-functional teams to implement design changes and meet customer

deadlines.

Design Engineer Intern

Apex Tech Centre, Pune

- Supported the design and development of automotive seating foam components using CATIA V5, considering component interfaces, packaging, and manufacturing requirements.
- Assisted in 3D modelling, design modifications, and engineering drawing updates for seating components based on design and customer requirements.

TECHNICAL SKILLS

CAD/CAE Tools: CATIA V5, Siemens NX, ANSYS, Abaqus

Programming Languages: Python (Pandas, NumPy, Scikit-learn), MATLAB, Fortran, SQL

ACHIEVEMENTS & RECOGNITION

Precisely Right Award - Team Category | Adient India Tech Center: Recognized for contributions to Stella Small, Strasbourg Foam Support, and Volvo V436 seating programs, including foam design, manufacturing tool development, and engineering support.

2nd Prize - ONYX 2017: Secured 2nd prize in a technical presentation event organized by the Department of Mechanical Engineering, MITAOE.

ADDITIONAL INFORMATION

Languages: English, German (A1), Hindi, Marathi

Interests: Cricket, Formula 1